Sumbot

A locally deployed, source-grounded weekly report agent

Track 2 - Development & Local Deployment of Private AI Agents

Project summary

Sumbot reads project materials, extracts facts with file-level attribution, drafts a concise weekly report, and reviews the draft for unsupported or outdated statements. It is designed for controlled local deployment through an OpenAI-compatible model endpoint.

Application scenarios

- Engineering teams consolidating meeting notes, daily logs, and previous reports.
- Project managers preparing repeatable weekly status updates with visible evidence sources.
- Small organizations keeping internal materials and generated reports inside their deployment.
- Demonstrators validating the full interface in mock mode before attaching a real local model.

Agent architecture

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User materials -> document normalization -> source-grounded Extractor -> constrained Writer -> independent Reviewer -> human approval and Markdown export
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The workflow is fixed and sequential. Prompts, intermediate facts, warnings, elapsed time, and failures remain inspectable. FastAPI provides the interface, SQLite stores local task state, and a one-worker queue prevents concurrent model jobs from competing for local inference resources.

Core capabilities

- Accept pasted text and .md, .txt, .docx, .pptx, .xlsx, and .pdf files.
- Retain source filenames in extracted facts and exported reports.
- Check the draft against original materials and extracted facts.
- Surface parsing timeouts, output truncation, interruption, and stage failures.
- Persist tasks locally and export completed reports as Markdown.

Model and algorithm

The model client uses configurable OpenAI chat completions. The recorded deployment uses qwen2.5-32b-it-Q4_K_M-LOT with Q4_K_M quantization, served through Lemonade 11.5.0 and its llama.cpp backend. Real calls use temperature 0.2 and independent output limits for extraction, writing, and review.

The Extractor merges duplicate facts and requires sources. The Writer may use extracted facts only. The Reviewer detects unsupported claims, time conflicts, plans presented as completed work, and truncation warnings.

Local deployment plan

1. Clone the repository in a Linux or Radeon Cloud environment.
2. Run bash scripts/run_app.sh to create the environment and install dependencies.
3. Start an OpenAI-compatible local inference server.

4. Set MOCK_LLM=false and the real LLM base URL, key, and model in ignored env.local.
5. Restart Sumbot, verify /health, and use the controlled Radeon Cloud tunnel if required.

AMD Radeon GPU use and inference optimization

AMD Radeon acceleration belongs to the local model-serving backend; Sumbot remains a portable workflow client. The recorded host exposes eight AMD Radeon PRO W7900D devices (Navi 31, gfx1100, PCI 1002:744b) and reports ROCm 7.2.1, AMDSMI 26.2.2, amdgpu 6.16.13, and HSA runtime 1.18. Device visibility does not establish that one request used all eight GPUs, so no multi-GPU utilization claim is made. This separation lets the backend select ROCm-compatible kernels, model formats, quantization, and memory settings without hardware-specific calls in business logic.

- Serialize model work to one job at a time.
- Bound output tokens independently for all three model stages.
- Limit parsed-document size and merge duplicate facts to reduce context pressure.
- Run inference in the background and report stage elapsed times.

Validation boundary: no latency, throughput, memory, or quality figure is claimed before a real Radeon run. The demo must show the GPU, runtime, actual model, MOCK_LLM=false, and observed timings.

Privacy and data handling

Uploaded binary files are parsed in memory and are not retained. Extracted text, task metadata, and results are persisted in local SQLite; exports are local Markdown files. Privacy depends on using a local endpoint and controlling tunnel access.

Current limitations

- Chinese report prompts only; filename-level rather than line-level attribution.
- No OCR for scanned PDFs or images; limited support for legacy/encrypted Office files and complex visual content.
- Reviewer findings assist but do not replace human factual approval.
- Running jobs do not resume after a service restart.
- Real Radeon performance remains to be verified in the final demonstration environment.

Deliverables

Complete source code, English README, this specification, an English supplementary poster, and a 3-5 minute video captured on the actual AMD Radeon environment.

Honest validation statement

The repository includes a deterministic mock backend for workflow testing. Mock output is not evidence of GPU execution. Only a real local endpoint with MOCK_LLM=false supports an AMD Radeon execution claim.